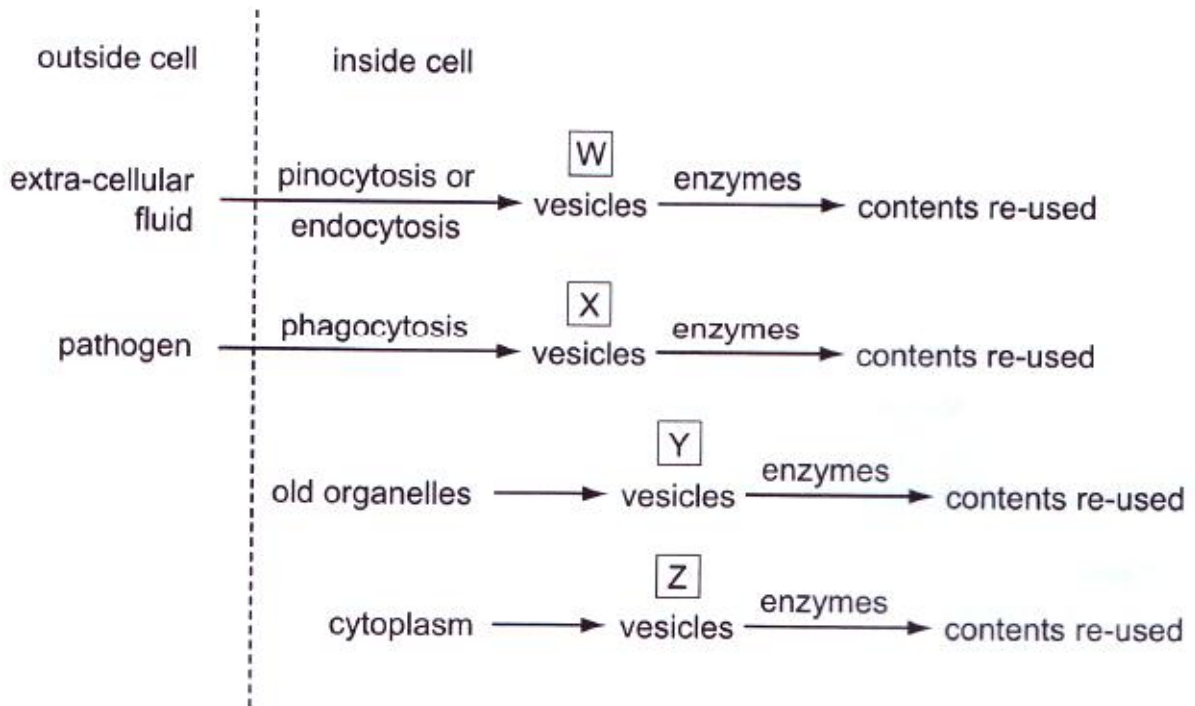


- 1 The flow chart shows processes which take place inside animal cells.

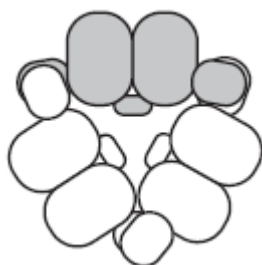
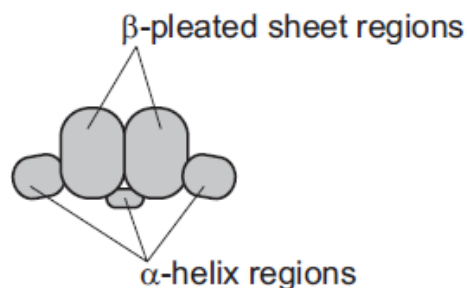


Which processes require the activity of lysosomes?

- A W, X, Y and Z
- B W and X only
- C X and Y only
- D Y and Z only

2 Approximately half of the total protein in a pea seed consists of the storage protein vicilin.

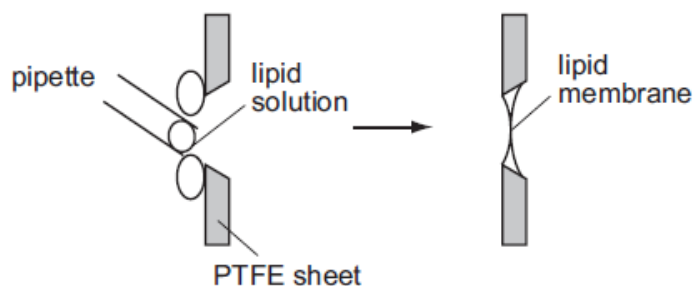
- Each molecule of vicilin is made up of three identical polypeptides.
- Each polypeptide is made up of two β -pleated sheet regions with linking α -helix regions, folded into the shape shown to the right.
- This allows the three polypeptides to pack together into a compact, flat storage molecule, as shown below.



Which row correctly describes the structure of vicillin?

	Primary structure	Secondary structure	Tertiary structure	Quaternary structure
A	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	association of three polypeptides	folding of each polypeptide
B	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide	association of three polypeptides
C	association of three polypeptides	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide
D	association of three polypeptides	amino acid sequence of one polypeptide	folding of each polypeptide	α -helix and β -pleated sheet regions of each polypeptide

- 3 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.



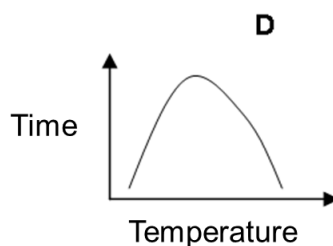
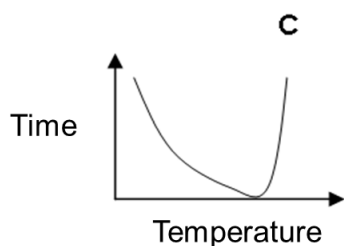
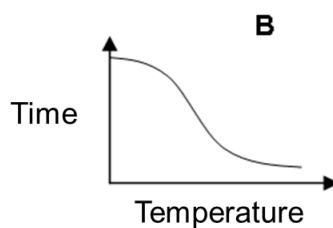
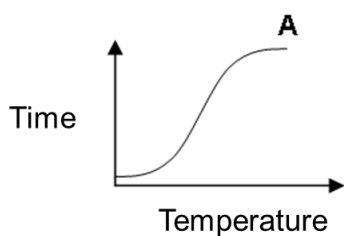
Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ . In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added, at which time current flowed.

Which statement is consistent with this information and your knowledge of membrane structure?

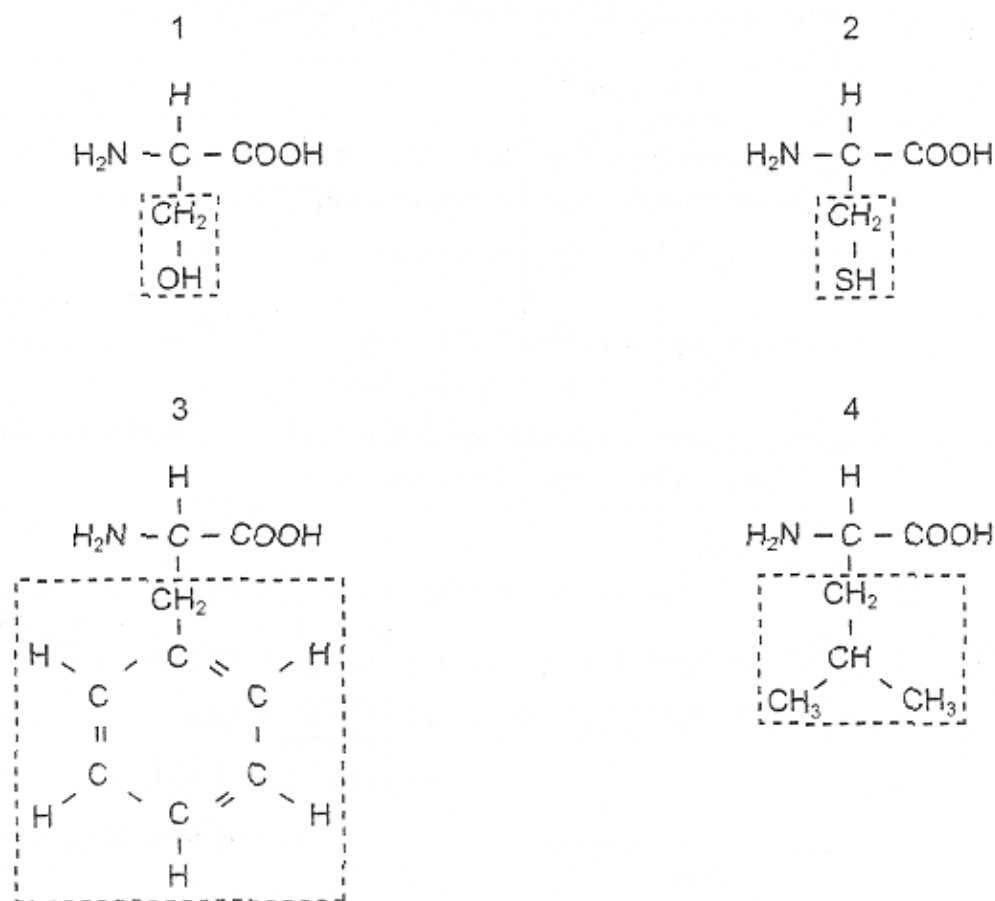
Gramicidin becomes incorporated into the membrane and is

- A a carbohydrate molecule found only on the outside of the membrane.
 - B a non-polar lipid which passes all the way through the membrane.
 - C a protein molecule with both hydrophilic and hydrophobic regions.
 - D a protein molecule which has only hydrophobic regions.
- 4 Iodine test gives a blue-black colour in the presence of starch. The time taken for the blue-black colour of a fixed amount of starch to disappear was recorded over a range of temperature.

Which graph shows the results of this investigation?



- 5 The diagram shows four different amino acids, each with a different R group (side chain).



Which amino acid could form a hydrophobic interaction between their R groups?

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 4 only
- D** 3 and 4 only

- 6 A living cell from the lung epithelium of a newt is shown at different stages of mitosis in the micrographs (A to F) below. Identify the correct sequence of these micrographs.

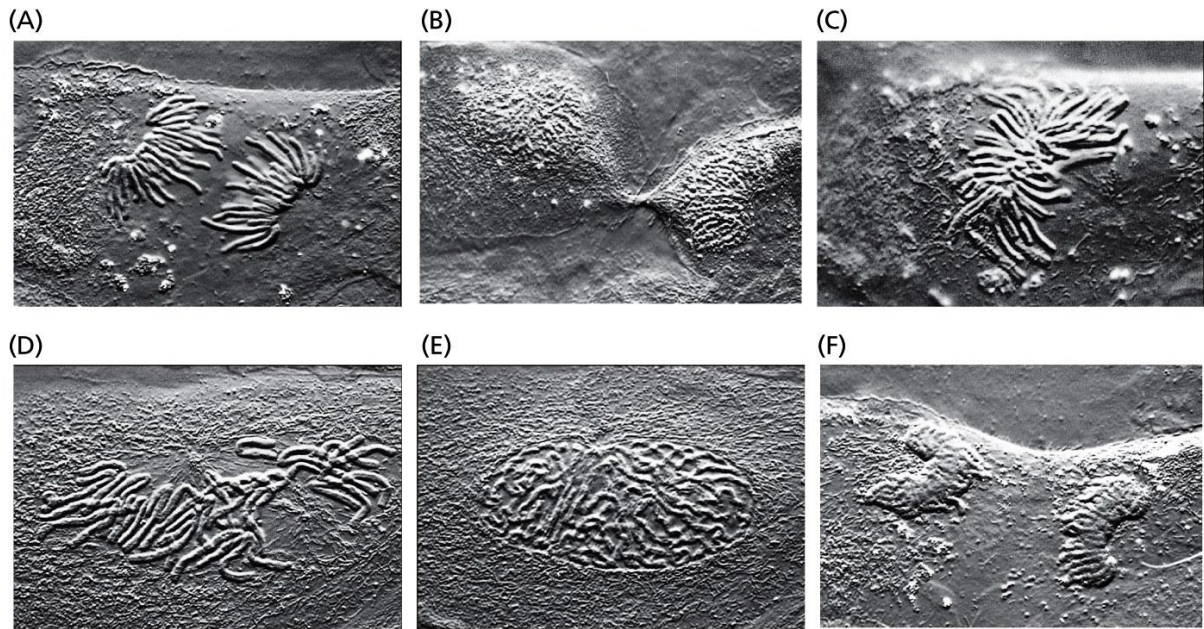


Figure 17-14 MBoc5: The Problems Book (© Garland Science 2008)

- A DECAFB
B EDCAFB
C ECDAFB
D EDCBAF

- 7 Which statements about the genetic code are correct?

1. The genetic code has redundancy and is degenerate.
2. There is only one codon for the amino acid methionine.
3. Codons act as 'stop' and 'start' signals during transcription and translation.
4. Prokaryotes generally use the same genetic code as eukaryotes.
5. mRNA codons have the same nucleotide sequence as DNA triplet codes.

- A 1, 2 and 3 B 1, 2 and 4 C 1, 3 and 5 D 2, 4 and 5

- 8 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced
B	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal

- 9 You have isolated an antibiotic, edeine, which inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to a translation mixture containing fully intact organelles, edeine stops haemoglobin translation within a few seconds.

Analysis of the edeine-inhibited mixture by centrifugation showed that no polyribosomes remained by the time protein synthesis had stopped. Instead all the mRNA had accumulated, together with small ribosomal subunits and initiator tRNAs.

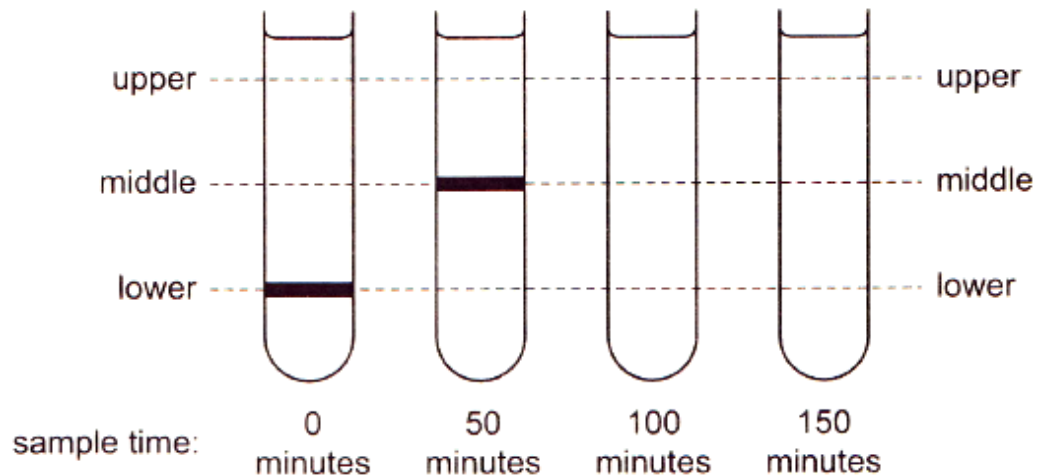
Which step in protein synthesis does edeine inhibit?

- A** It prevents the formation of the translation initiation complex.
- B** It inhibits the binding of aminoacyl-tRNA to the A site in the ribosome.
- C** It blocks the translocation of peptidyl-tRNA from the A site to the P site of the ribosome.
- D** It interferes with chain termination and release of peptide.

- 10** The generation time of *Escherichia coli* at 36 °C is 50 minutes. *E. coli* were grown on a medium containing the heavy isotope of nitrogen, ^{15}N , so that the DNA of all the cells was labelled with ^{15}N . The cells were divided into two groups. In the first group DNA was extracted from the cells forming the sample at 0 minutes.

Cells in the second group were transferred to a culture medium containing the normal isotope of nitrogen, ^{14}N . Samples of cells were removed from this second group at 50-minute intervals and their DNA extracted. The extracted DNA from each sample was centrifuged in density gradient tubes of caesium chloride.

The diagram shows the possible positions (upper, middle and lower) of bands of DNA in these tubes. The actual positions of bands in the first two samples are shown.



Which proportion of the DNA of the sample taken at 150 minutes will be at the upper position?

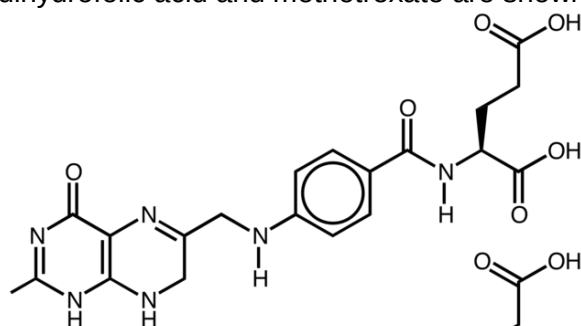
- A** 25% **B** 50% **C** 75% **D** 100%

- 11 Burkitt's lymphoma is a cancer of the lymphatic system associated with Epstein-Barr virus (EBV). Treatment of Burkitt's lymphoma involves chemotherapy that uses a combination of cytotoxic anti-cancer drugs.

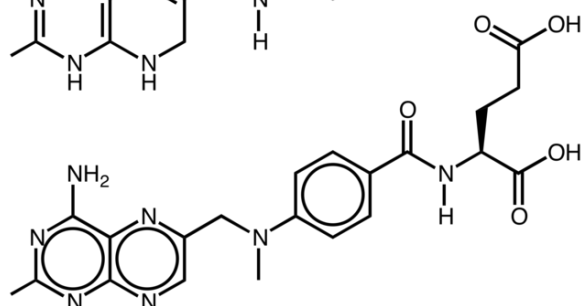
One of these cytotoxic drugs is methotrexate that acts on dihydrofolate reductase (DHFR), an enzyme that converts dihydrofolic acid to tetrahydrofolic acid.

The chemical structures of dihydrofolic acid and methotrexate are shown in the figures below.

Dihydrofolic acid

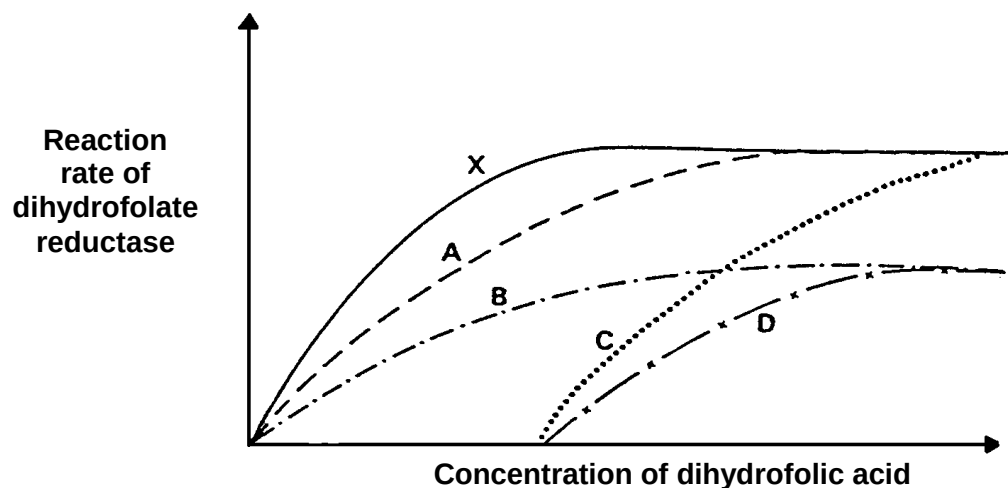


Methotrexate

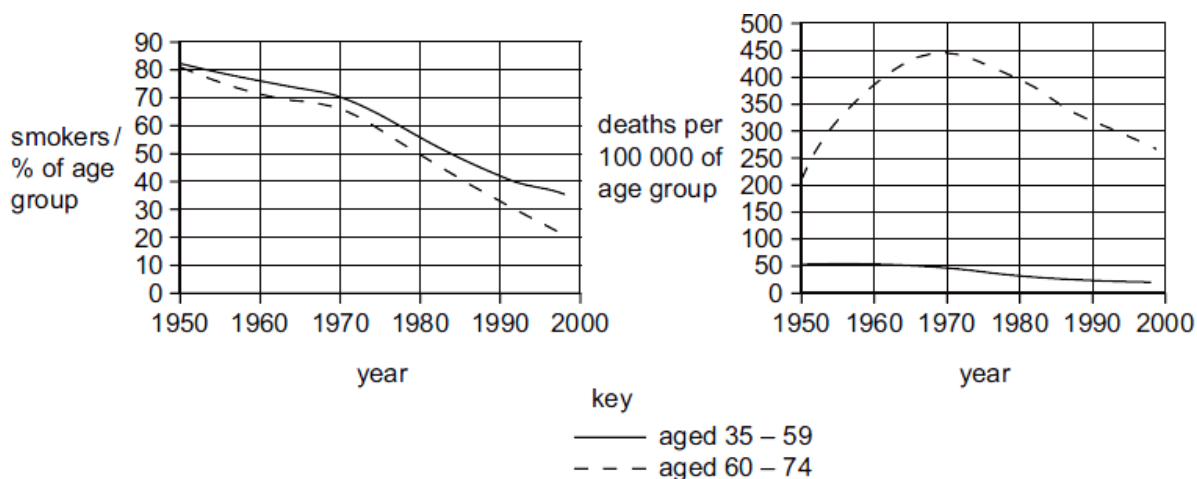


Graph **X** below represents the relationship between the reaction rate of dihydrofolate reductase and concentration of dihydrofolic acid under optimal conditions.

Which of the following graphs labelled **A** to **D** represents the altered rate of reaction of dihydrofolate reductase under the action of the drug methotrexate?



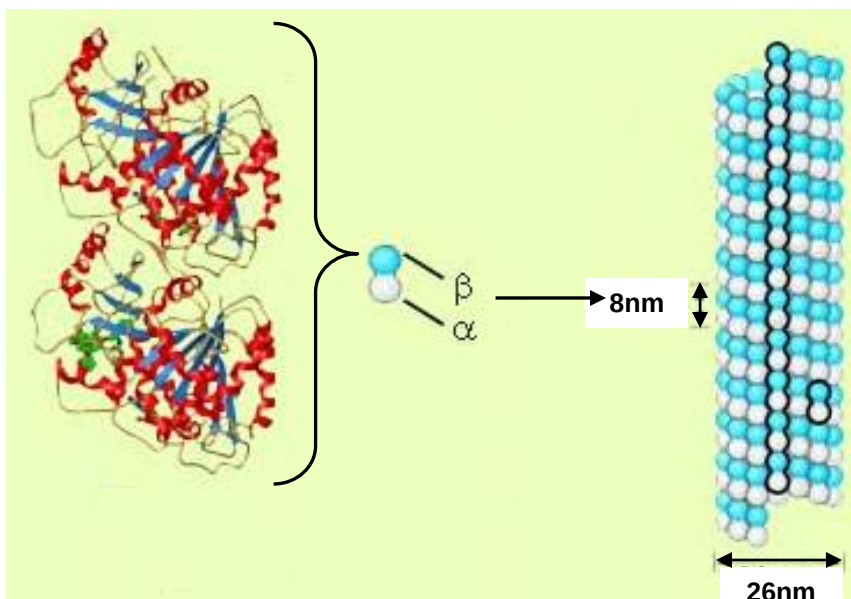
- 12** Some studies suggest that smoking increases the risk of developing lung cancer. The two graphs show the percentage of smokers and the deaths from lung cancer in men of two age groups between 1950 and 1998.



Which statement is not supported by the data in the graphs?

- A** Deaths from lung cancer in men aged 35-59 decreased by 50% over the period of the study.
- B** Deaths from lung cancer in men aged 60-74 increased up to 1970.
- C** The data for men aged 60-74 between 1950 to 1970 suggests that lung cancer takes up to 20 years to develop.
- D** The number of men aged 35-59 who were smokers decreased by approximately 60 % over the period of the study.
- 13** One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long. What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?
- A** 43 nm
- B** 66 nm
- C** 132 nm
- D** 387 nm

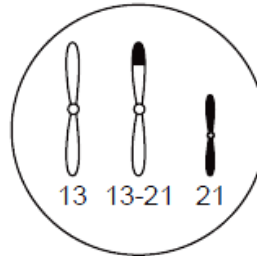
- 14 Spindle fibres involved in mitosis are formed by aggregation of the protein tubulin. The diagram below shows the dimerisation of tubulin subunits α and β , and the aggregation of the tubulin dimers to form microtubules.



Which of the following correctly describe the tubulin protein?

1. The structure of the tubulin subunits is maintained by interactions between R groups.
 2. Microtubules exhibit the secondary protein structure as they are long helices.
 3. Tubulin subunits are secondary proteins as they contain α -helices.
 4. The tubulin protein consists of both covalent and non-covalent bonds.
 5. The tubulin dimers are tertiary protein structures, while the microtubules are quaternary structures.
- A** 1 and 4 only
- B** 2 and 4 only
- C** 1, 2 and 3 only
- D** 1, 4 and 5 only

- 15** Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21. The diagram shows chromosomes 13 and 21 in the nucleus of a diploid (2n) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



Which is not a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$; normal phenotype
B	13-21	$2n = 45$; normal phenotype
C	13-21 and 21	$2n = 46$; Down's syndrome
D	13-21 and 21	$2n = 47$; Down's syndrome

- 16** The table shows the results of an early investigation into the genetic control of phenotypic variation. The dry masses of 5493 bean seeds collected from many plants were classified into nine categories.

mass of bean / mg	51-150	151-250	251-350	351-450	451-550	551-650	651-750	751-850	851-950
number of beans	5	38	370	1676	2255	928	187	32	2

Which statement correctly describes these data and could account for the variation shown?

- A** The phenotypic variation is continuous and could be the result of two non-linked genes acting on their own.
- B** The phenotypic variation is continuous and could be the result of several non-linked genes acting on their own.
- C** The phenotypic variation is discontinuous and could be the result of two linked genes acting on their own.
- D** The phenotypic variation is discontinuous and could be the result of several linked genes acting on their own.

- 17 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

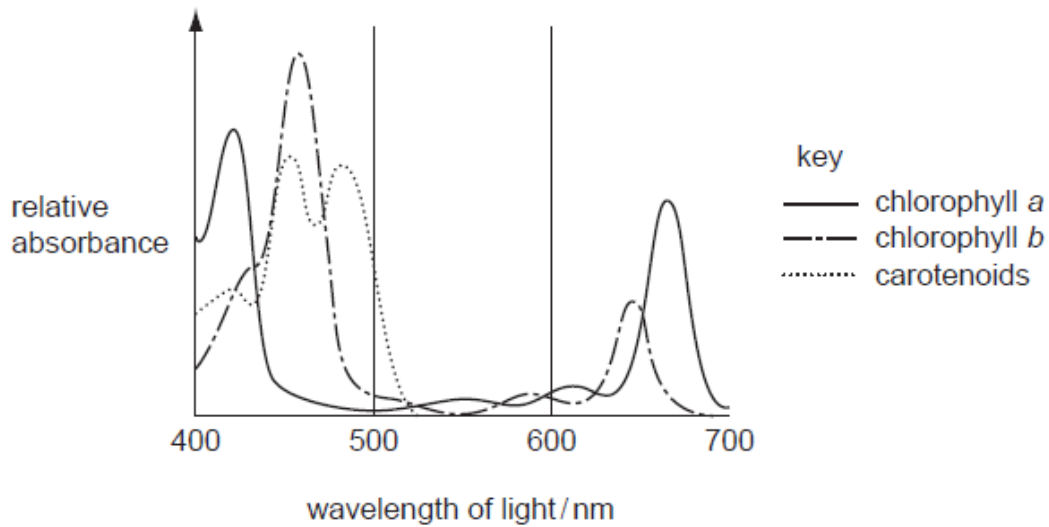
What explains the inheritance of the range of phenotypes shown by these crosses?

- A two genes, each with a dominant and recessive allele
 - B sex linkage of the allele for grey coat colour
 - C one gene with multiple alleles
 - D one gene with a pair of co-dominant alleles
- 18 Albino horses have recessive alleles which cannot be transcribed to produce pigmentation. Geneticists studying horses have found a different allele of the same gene which predisposes carriers to premature aging of the hair follicles. This allele is dominant, so, even when born with pigmented hair, the coats of horses with this allele will gradually turn white over the first 6 to 8 years of life.

Which **white** female horse would be best to use in crosses to determine the genotype of a ten-year-old, white male horse?

- A an unrelated, ten-year-old female
- B an unrelated, three-year-old-female
- C a sister of the male
- D the male's mother

19 The graph shows the absorption spectra of some pigments found in chloroplasts.

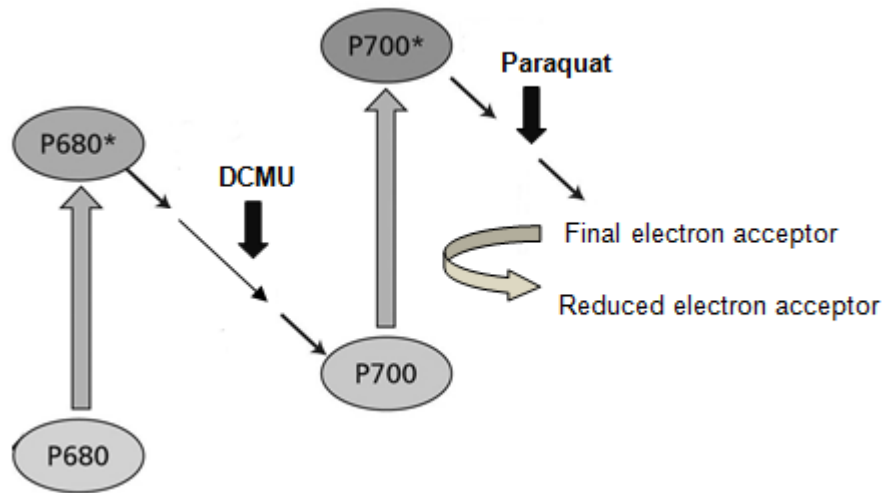


Which statements are correct?

1. Having several pigments, rather than one, increases the efficiency of photosynthesis.
2. Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
3. Photosynthesis will be fastest in light at the red end of the spectrum, as red light has higher energy than blue light.
4. Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.

- A** 1 and 2 only
- B** 1 and 3 only
- C** 1, 2 and 4 only
- D** 2 and 4 only

- 20 The diagram shows the electron transport chain embedded in thylakoid membranes of chloroplasts.

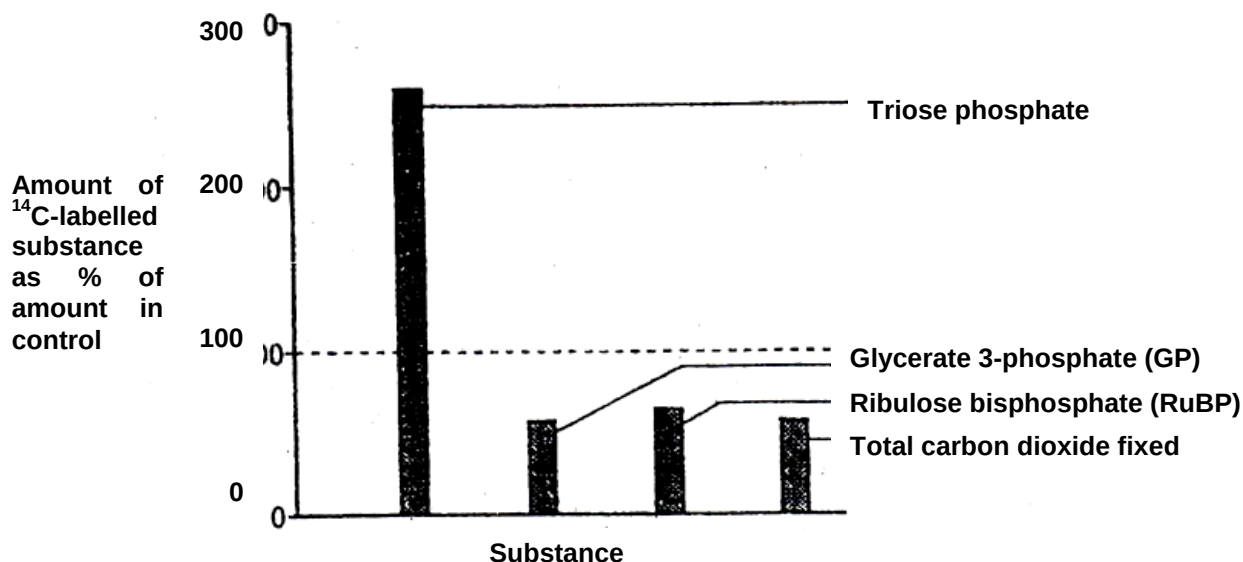


DCMU and paraquat are inhibitors of electron transport chain at the positions indicated in the diagram. These two inhibitors are added separately to isolated chloroplasts.

Which of the following correctly represents the results in the presence of DCMU and paraquat?

	DCMU only			Paraquat only		
	NADPH	oxygen	ATP	NADPH	oxygen	ATP
A	Present	Absent	Present	Absent	Absent	Present
B	Absent	Present	Absent	Absent	Present	Present
C	Absent	Present	Present	Present	Present	Absent
D	Absent	Absent	Absent	Absent	Absent	Present

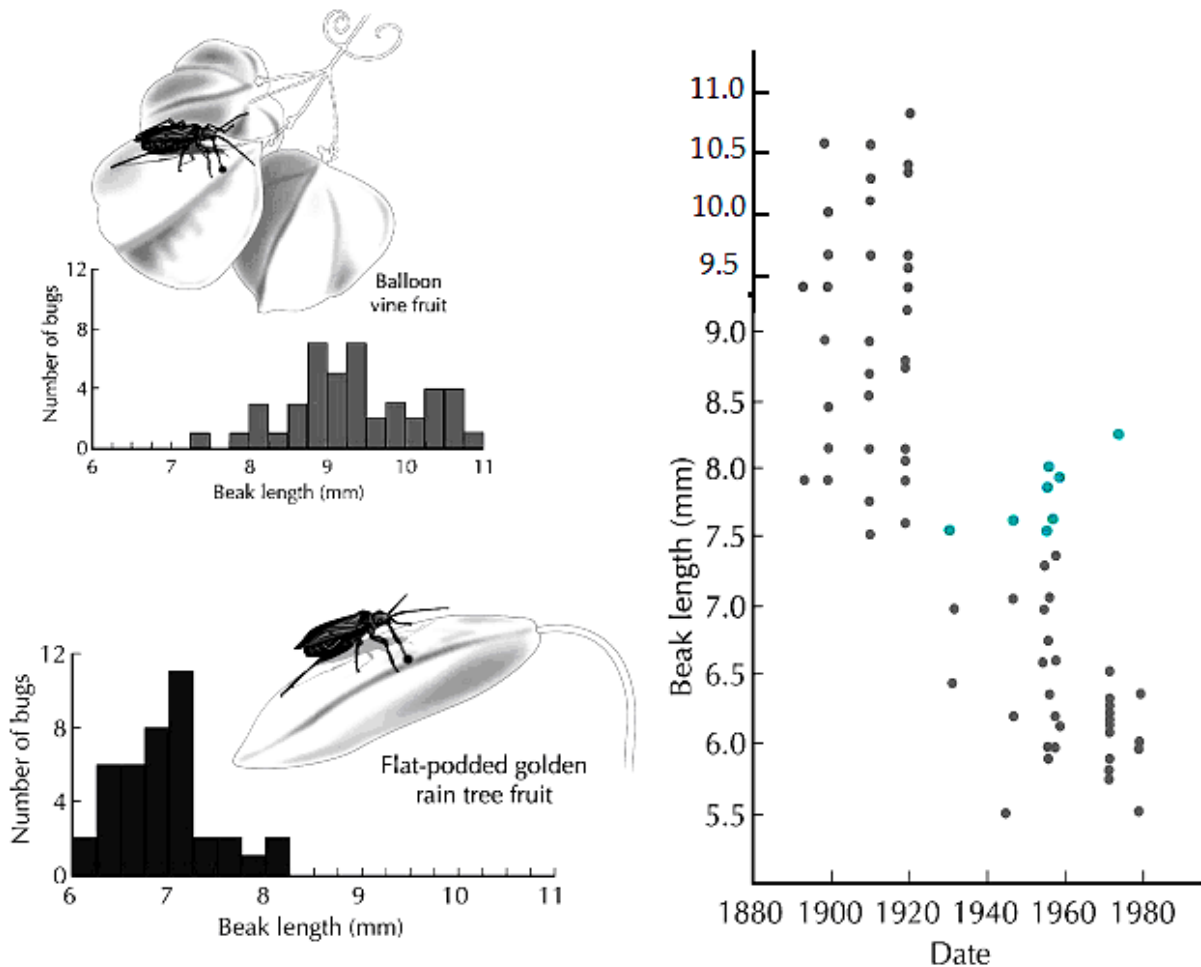
- 21 An illuminated suspension of unicellular alga *Chlorella* in cadmium chloride was treated with carbon dioxide labelled with ^{14}C . The light was switched off and the amount of radioactivity present in some intermediates was determined after 10 minutes in the dark. A control suspension of cells without cadmium chloride was treated in exactly the same way. The bar chart below shows the amount of radioactivity in these intermediates in the cells from the cadmium chloride solution compared with the control cells.



From the above data, which of the following deductions can be made?

- A Cadmium chloride causes triose phosphate to accumulate as it prevents triose phosphate from being converted to hexose sugars.
- B Cadmium chloride prevents the regeneration of RuBP by inhibiting the enzyme at the stage just after triose phosphate.
- C Cadmium chloride increases the rate at which glycerate-3-phosphate is being reduced to triose phosphate.
- D Cadmium chloride decreases the total amount of carbon dioxide fixed by RuBP by inhibiting RuBP carboxylase oxygenase.

- 22 Soapberry bugs in America originally fed on the native balloon vine using their sharp beaks to penetrate the fruit. In the 1920s, the flat-podded golden rain tree was introduced from Asia. This has thinner-skinned fruit.



What can you conclude from the data presented above?

1. Soapberry bugs with shorter beaks are being selected for.
 2. Soapberry bugs evolved shorter beaks after they switched to feed on this new host plant.
 3. Soapberry bugs had a preference for the introduced flat-podded golden rain tree.
 4. Soapberry bugs population contains the allele for short beaks since 1900.
- A 1 only
- B 1 and 2 only
- C 1, 2 and 3 only
- D 2, 3 and 4 only

- 23** Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome C. Cytochrome C is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that

- all living organisms are related
 - there is a single, rather than a multiple, origin of life?
1. The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
 2. The sequence of amino acids in cytochrome C is similar in organisms that are from similar environments or with similar metabolic demands.
 3. The majority of organisms have the same, or similar, amino acid sequences for cytochrome C.
 4. When transferred into a very dissimilar organism, a gene coding for cytochrome C will lead to the expression of a protein that will function in the other organism.

- A** 1 and 2 only
- B** 2 and 3 only
- C** 3 and 4 only
- D** 1, 3 and 4 only

- 24** The human genome project has identified and mapped the genes on the human chromosomes. This is allowing scientists to identify specific, faulty genes which contribute to inherited conditions.

This is useful in many ways, for example

1. carriers of faulty genes can be advised about changes in lifestyle to minimise risks.
2. carriers of faulty genes can be identified and informed of their risk status.
3. diagnostic tests can be developed to identify carriers of faulty genes.
4. drugs can be developed to block the action of problem genes.
5. embryos can be screened to avoid the birth of affected children.
6. employers can take account of the genetic predisposition of the employees.

Which two uses arise directly from the information provided by the project?

- A** 1 and 2 **B** 2 and 5 **C** 3 and 4 **D** 5 and 6

- 25** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

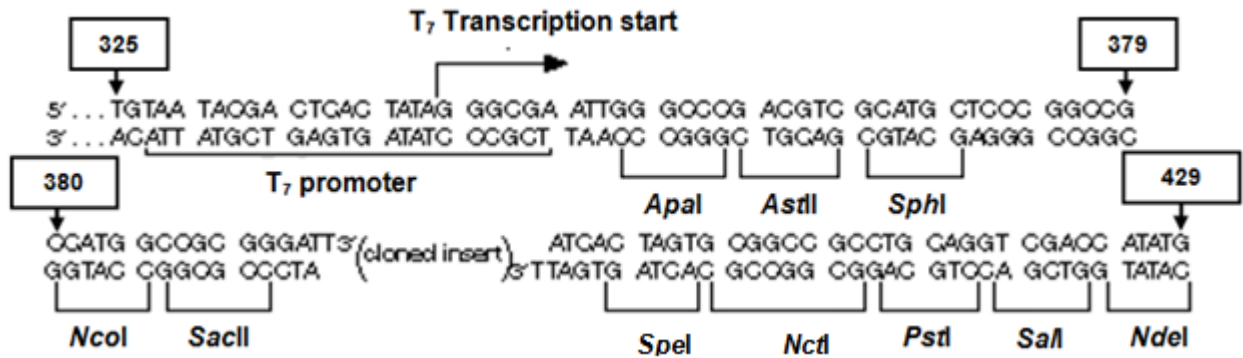
Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statement correctly describes the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

1. *Wolbachia* acts as a selective agent.
 2. The selective killing of male embryos is an example of artificial selection.
 3. When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
 4. All male embryos that carry the dominant alleles of the suppressor gene pass that allele to their offspring.
 5. The frequency of the dominant allele of the suppressor gene rises in the butterfly population.
- A** 1 and 4 only
- B** 1, 3 and 5 only
- C** 2 and 3 only
- D** 2, 4 and 5 only

Questions **26** and **27** refer to the information below.

pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, *Tet^R* and *lacZ*. The figure below shows a portion of the multiple cloning site (MCS) of the cloning vector. It is located within the *lacZ* gene.

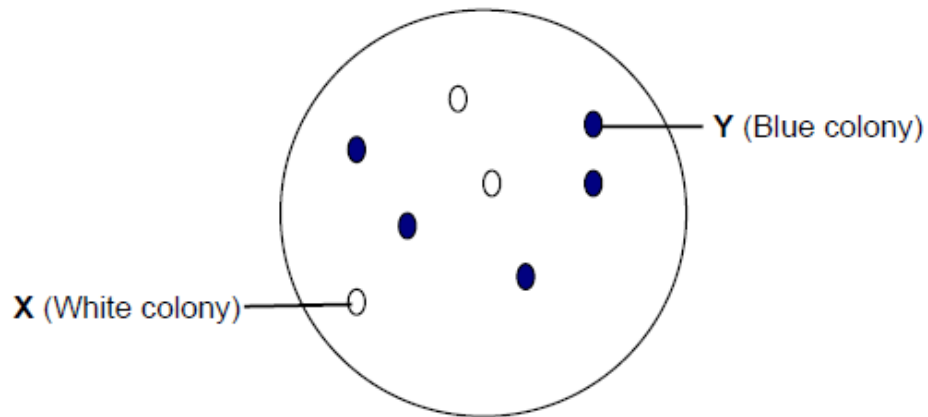


Note: The number in each box represents the nucleotide position within the plasmid.

- 26** Which of the following is the resulting number of fragments when pGEM-T is digested by the respective enzymes?

	SacII only	SacII and <i>SpeI</i>
A	2	3
B	2	2
C	1	3
D	1	2

- 27** Using *Sac*II as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS as shown in the figure on page 20 and the recombinant plasmid is transformed into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



Colonies **X** and **Y** are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.25kb and 0.34kb fragments

28 In obtaining the gene that codes for human actin, reverse transcriptase was used.

Why is reverse transcriptase **not** used for obtaining the gene that produces a human sodium ion channel protein?

- A** Reverse transcriptase is a viral enzyme, and in the current environment of suspicion of viruses, it is not possible to use such enzymes.
- B** The amino acid sequence for this protein is not known, so this is a gene that has not yet been located by the human genome project.
- C** The antibiotic resistance genes transferred by reverse transcriptase with the insulin gene may be transferred to pathogens.
- D** The sodium ion channel protein gene is expressed at low levels in human cells, so the mRNA produced is swamped by other mRNA.

29 One type of GM tilapia that is close to commercial consideration is a hybrid of two tilapia species, with transgene consisting of a tilapia growth hormone cDNA spliced to a viral promoter.

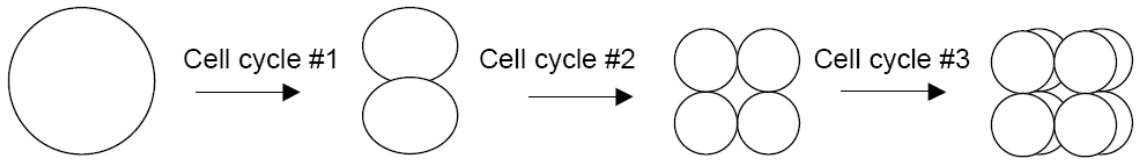
Consumers and critics have expressed some concerns about the production of GM tilapia.

1. excessive production of growth hormone in the tilapia
2. GM fish gaining a mating advantage over wild tilapia should they escape
3. production of foreign protein in tilapia
4. undesirable effects of transgene

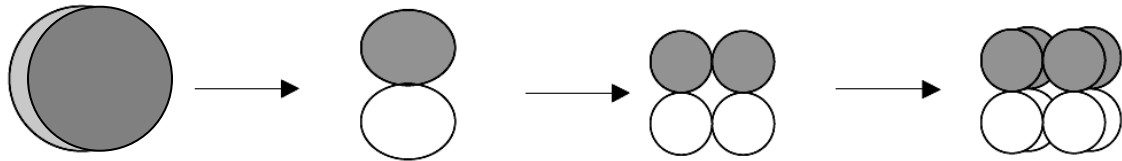
Which of their concerns is / are not valid?

- A** 1 and 2 only
- B** 2 and 3 only
- C** 3 only
- D** 4 only

- 30** When a fertilized egg begins development, the outcome of the first three cell cycles is shown as follows:



The localization of a specific protein during each of the above four early developmental stages is examined. This protein is required for the formation of ectoderm. The location of the protein is indicated by grey shading.



At the four-cell stage, are the four cells equivalent in terms of the different fates their daughter cells can take on, and why?

	Yes or No to the four-cell equivalence in terms of their fates	Justification
A	No	The protein required for ectoderm formation is found only in the upper two cells; therefore, only these two cells can become ectoderm. The lower two cells can continue to differentiate to become mesoderm or endoderm at this point, but cannot become ectoderm.
B	No	The protein required for ectoderm formation is found only in the upper two cells. These proteins can be transported to the other two lower cells such that the mesoderm or endoderm can be formed.
C	Yes	The protein required for ectoderm is needed for differentiation of the other germ layers since the ectodermal cells still remain totipotent and can take on any developmental pathway.
D	Yes	The protein required for ectoderm is needed only by the upper two cells since the lower two cells can differentiate to become mesoderm or endoderm at this point.

2013 H1 Biology Preliminary Examination Paper 1 ANSWER KEY

1	A	2	B	3	C	4	C	5	D
6	B	7	B	8	D	9	A	10	C
11	A	12	D	13	C	14	A	15	D
16	B	17	C	18	A	19	C	20	D
21	B	22	A	23	C	24	C	25	B
26	D	27	A	28	D	29	C	30	A